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# Activation Function

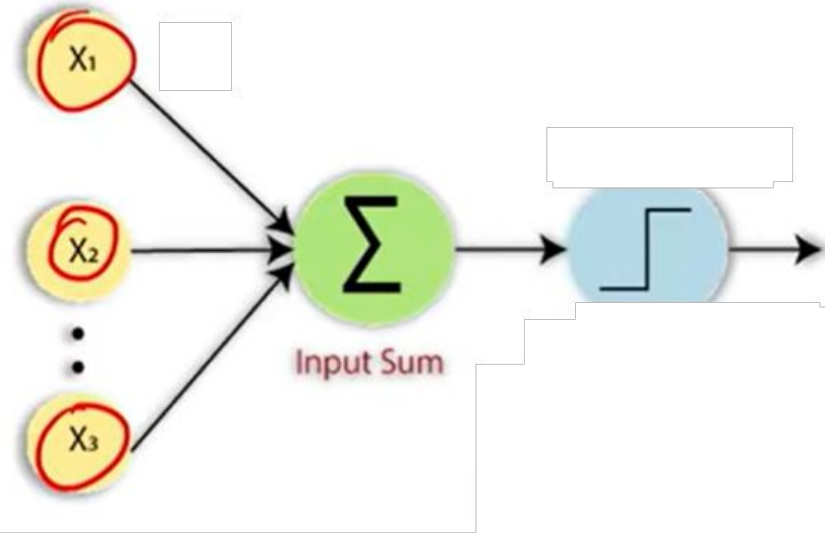
— Dr. Anuradha Yenikar —

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# Activation Function

**An Activation Function** decides whether a neuron should be activated or not. This means that it will decide whether the neuron's input to the network is important or not in the process of prediction using simpler mathematical operations.



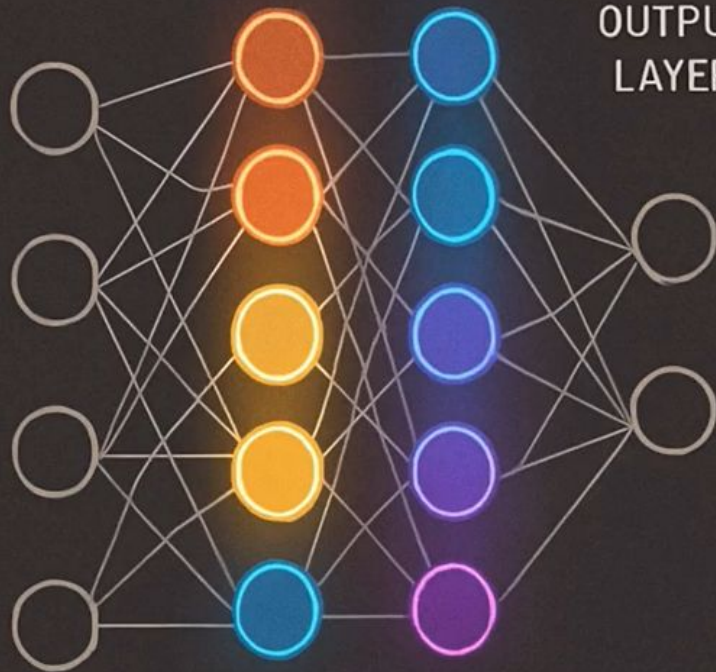
Output is Binary / Regression in range  $-\infty$  to  $+\infty$



INPUT  
LAYER

HIDDEN LAYERS

OUTPUT  
LAYER



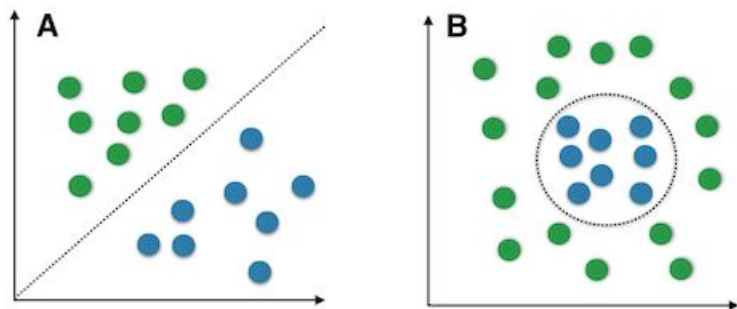
# Activation Function

➤ The activation function is categorized

into two main parts :

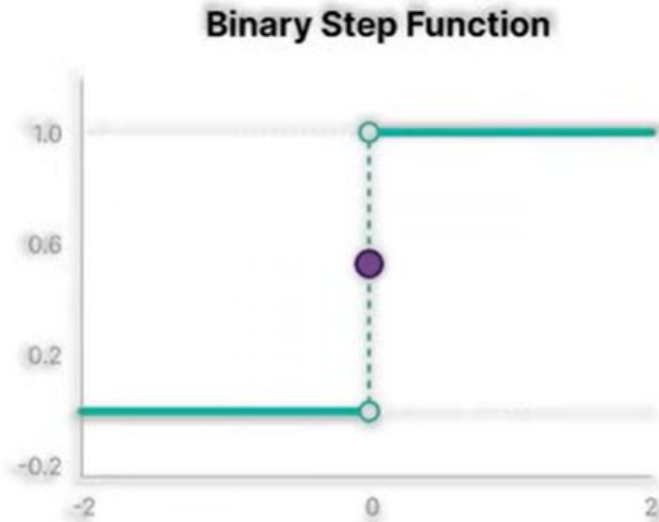
- Binary Step Function
- Linear Activation Function
- Non-Linear Activation Function

Linear vs. nonlinear problems



# Binary Step Function

**Binary Step Function :** Binary step function depends on a threshold value that decides whether a neuron should be activated or not.



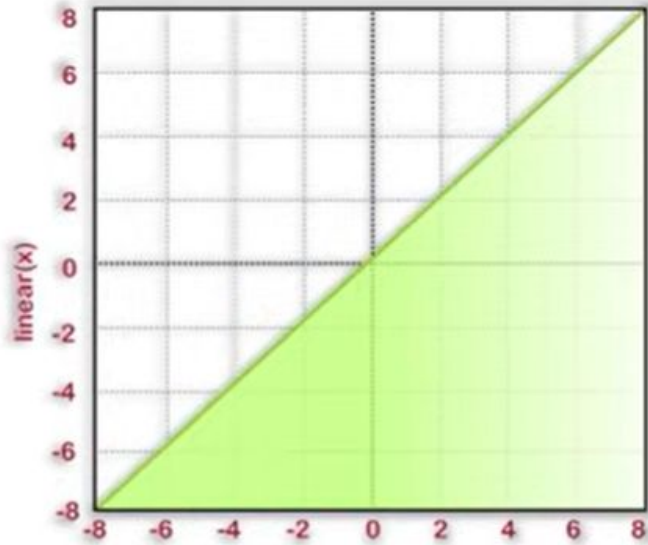
*Binary step*

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$

$$\hat{Y} = X_1W_1 + X_2W_2 + b$$

# Linear Activation Function

**Linear Activation Function :** The output of functions is not restricted in between any range.  
Its range is specified from  $-\infty$  to  $\infty$ .



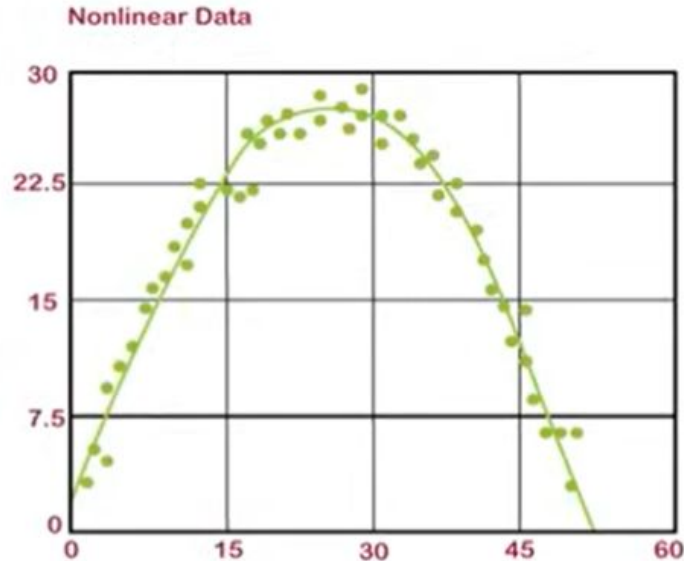
*Linear*

$$f(x) = x$$

$$Y = X$$

# Non-linear activation function

**Non-linear function:** These are one of the most widely used activation function. It helps the model in generalizing and adapting any sort of data in order to perform correct differentiation among the output.



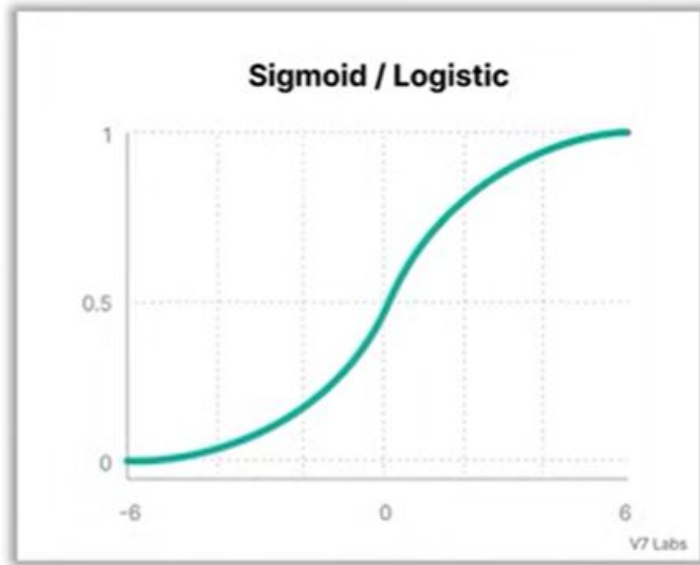
# Non-linear activation function

## ➤ Non-Linear Neural Networks Activation Functions

- ▪ Sigmoid / Logistic Activation Function
- Tanh Function (Hyperbolic Tangent)
- ReLU Function
- Softmax Function

# Sigmoid / Logistic Activation Function

• Sigmoid / Logistic Activation Function :

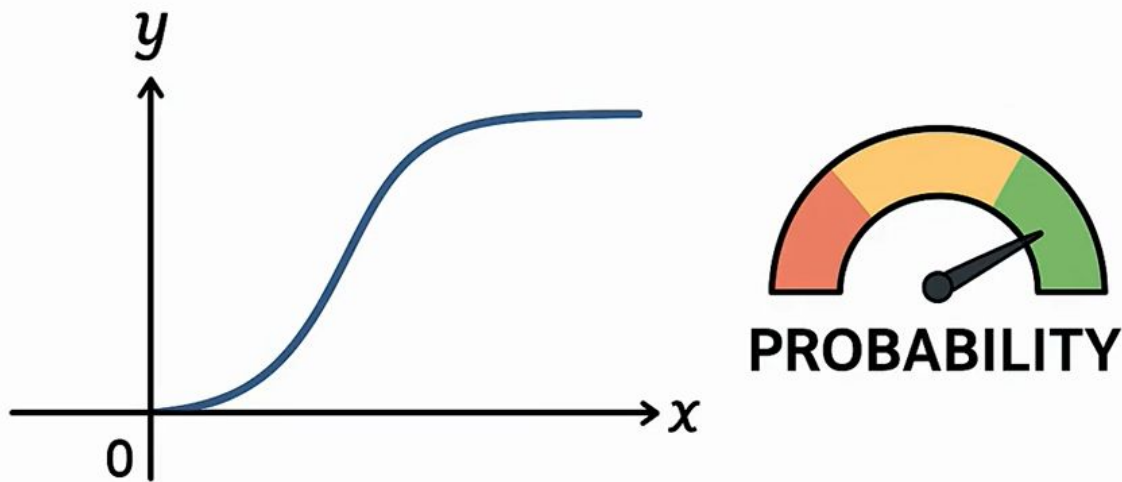


$$f(x) = \frac{1}{1 + e^{-x}}$$

**Output - Binary (0 / 1)**

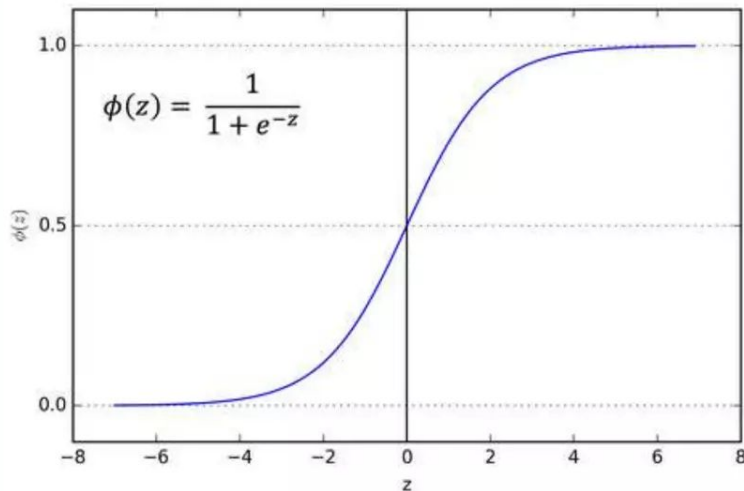
**It is differentiable activation function. Means we can use in Gradient Descent.**

# SIGMOID



**Good for probabilities,  
but suffers from vanishing  
gradients**

# 1. SIGMOID ACTIVATION FUNCTION

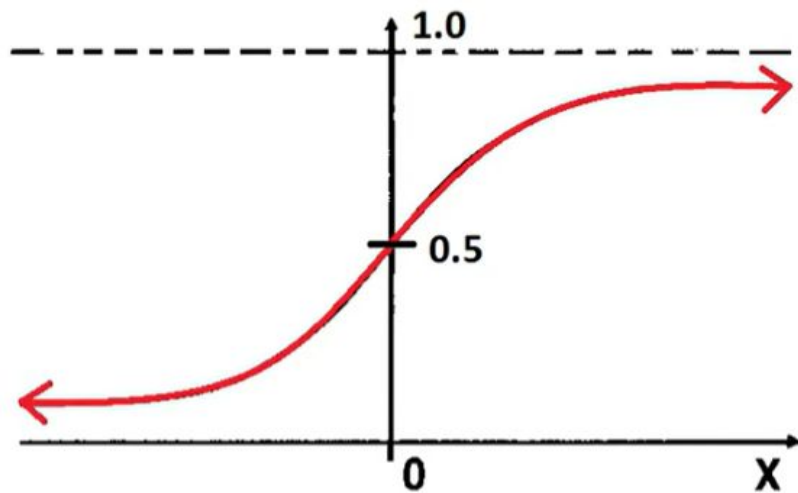


Sigmoidal functions are frequently used in [machine learning](#), specifically in the testing of artificial [neural networks](#), as a way of understanding the output of a node or “[neuron](#).”

A sigmoid function is a type of [activation function](#), and more specifically defined as a squashing function. Squashing functions limit the output to a range between 0 and 1.

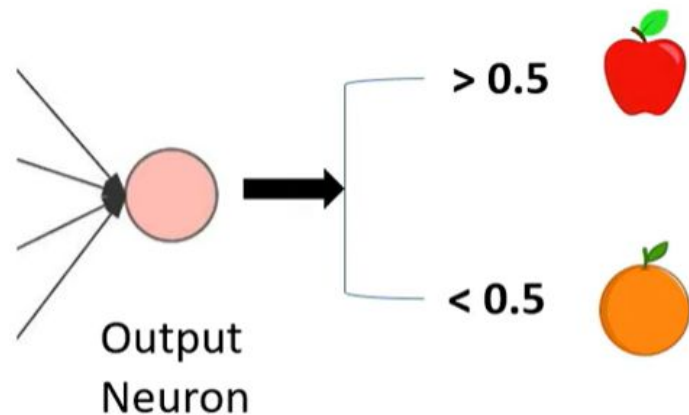
# Sigmoid Function

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*Sigmoid*

$$f(x) = \frac{1}{1+e^{-x}}$$





# Pros And Cons of Sigmoid Activation function

## Pros

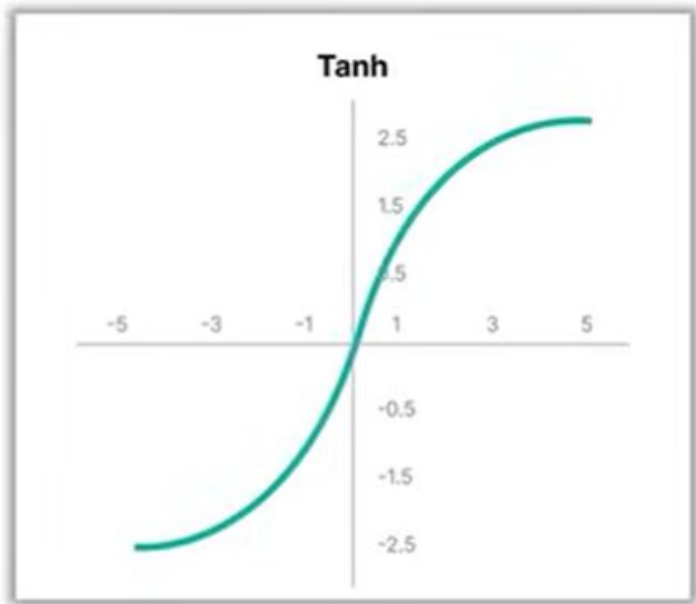
1. The performance of Binary Classification is very well as compare to other activation function.
2. Clear predictions, i.e very close to 1 or 0.

## Cons

1. The calculation in sigmoid function is complex.
2. It is not useful in multiclass classification .
3. For negative values of x-axis gives 0.
4. It become constant and gives 1 for any high positive values.
5. Function output is not zero-centered

# Tanh Activation Function

Tanh Function (Hyperbolic Tangent) :

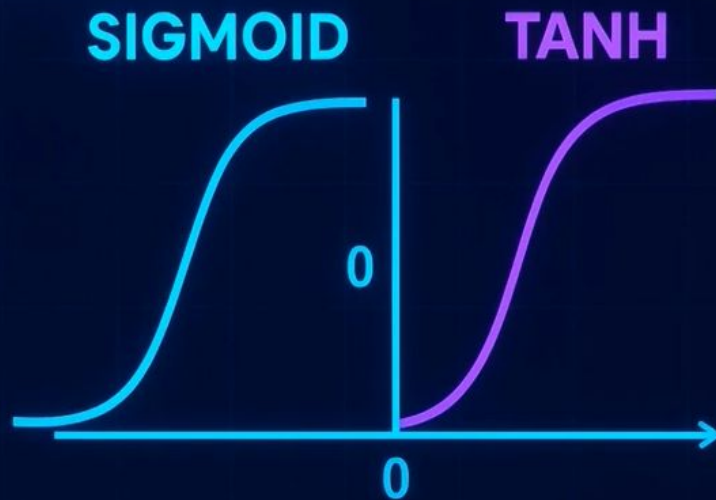
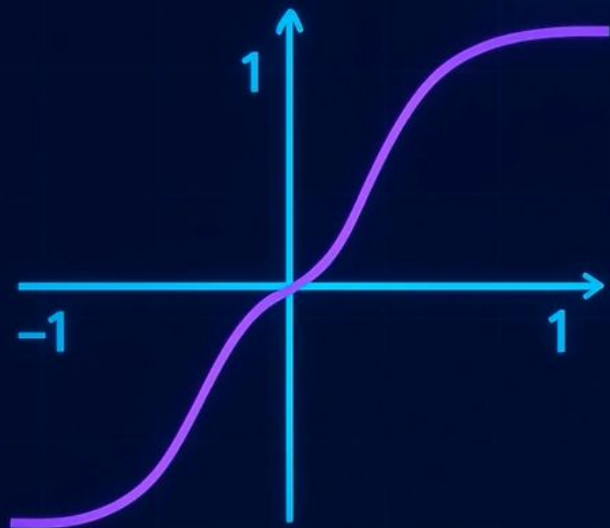


$$f(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$$

Used in RNN majorly. Output : -1 to +1

It is zero centered. Means we can use in Gradient Descent.

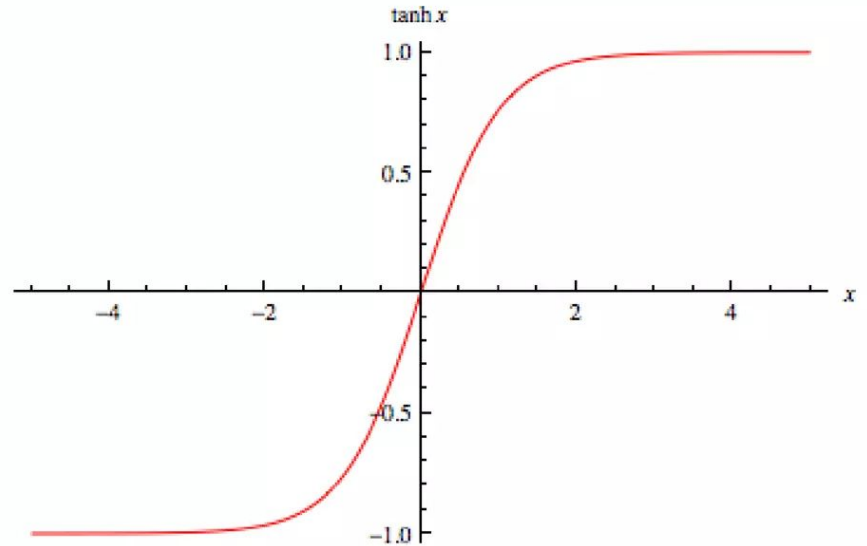
# TANH



**BETTER THAN SIGMOID,  
BUT NOT PERFECT**

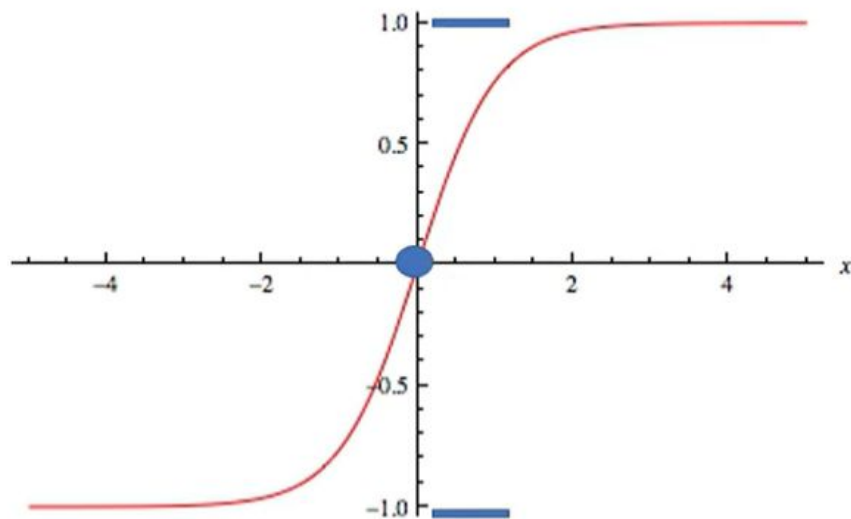
# Hypertangent activation Function

This **function** is easily defined as the ratio between the **hyperbolic** sine and the cosine **functions**



# Tanh Function

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$$\tanh = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

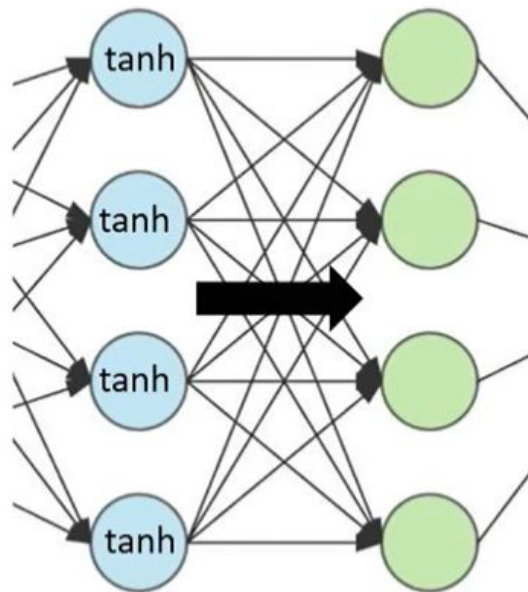
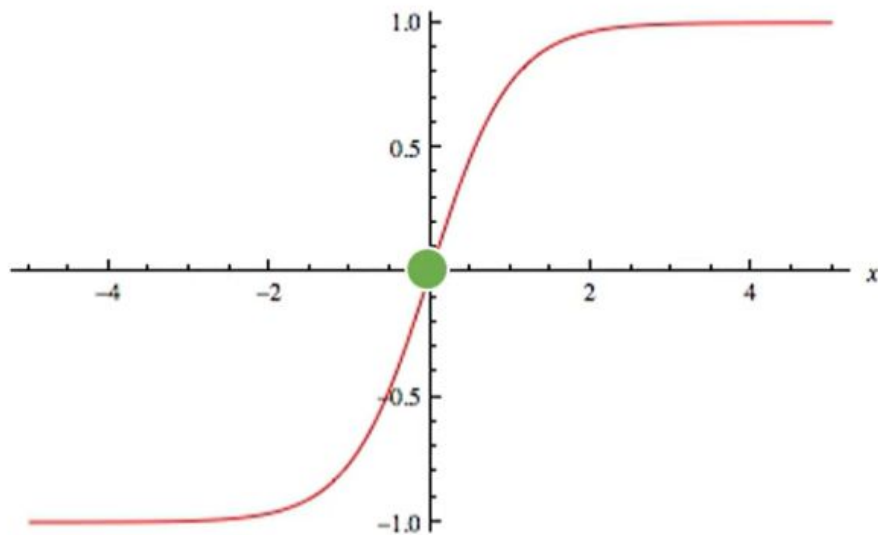
$$W = W - \alpha * \frac{\partial \text{cost}}{\partial W}$$

We need to take  
derivative of activation  
functions

$$\frac{\partial \tanh}{\partial x}$$

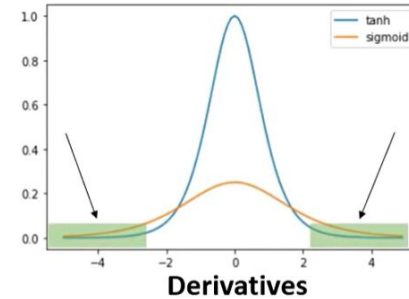
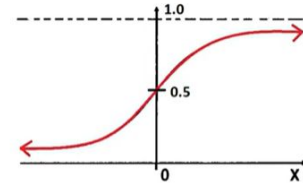
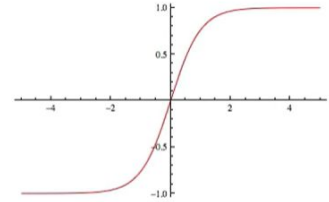
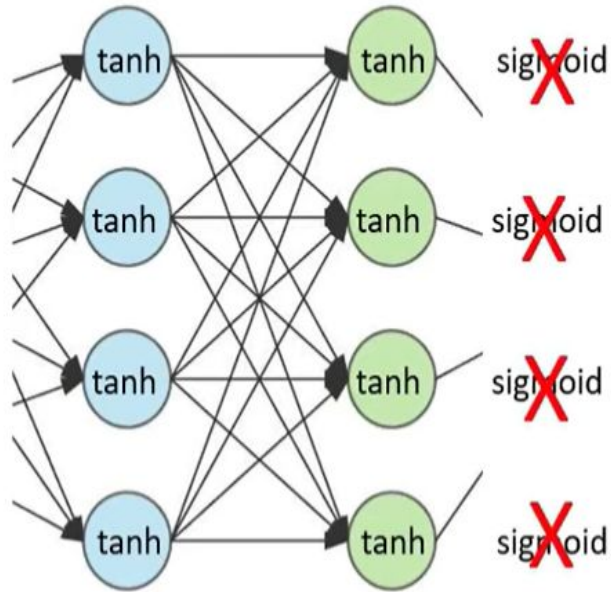
# Tanh Function

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*Data is more centered towards 0. Thus, learning is easier.*

# Tanh Function



Smaller value of derivative leads to very slow learning

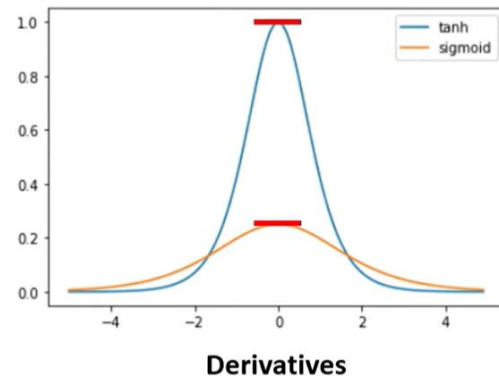
# Pros And Cons of Tanh Activation function

## Pros

1. The gradient is stronger for tanh than sigmoid ( derivatives are steeper).
2. The output interval of tanh is 1, and the whole function is 0-centric, which is better than sigmod

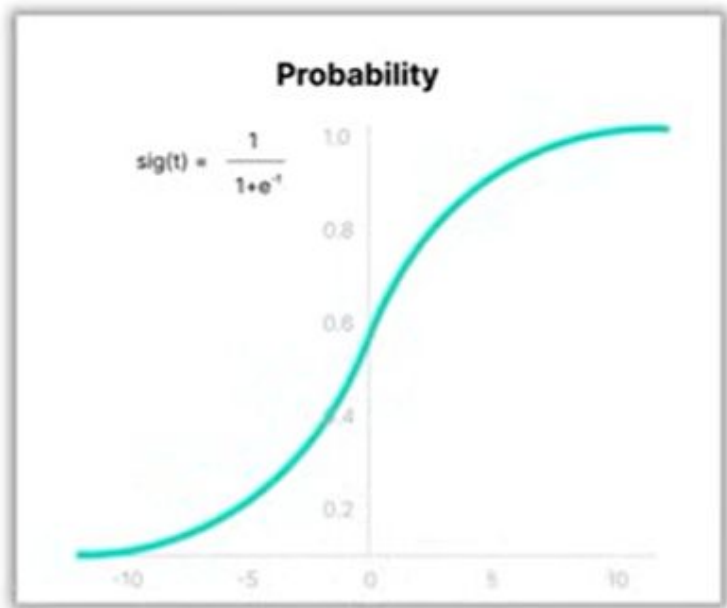
## Cons

1. Tanh also has the vanishing gradient problem.



# Softmax Activation Function

SoftMax Function :



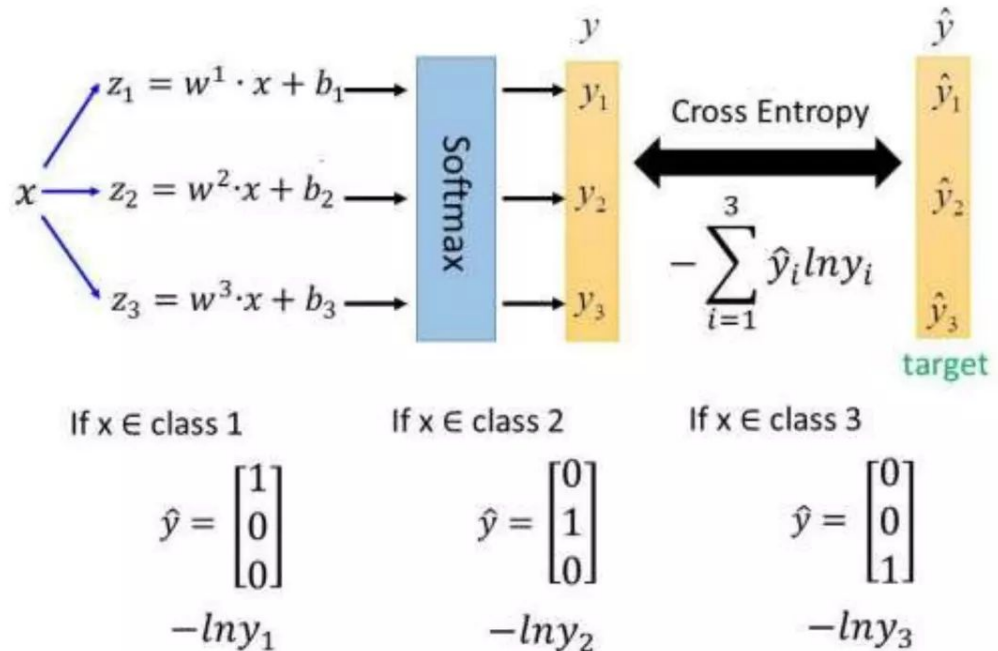
$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

When output is categorical, softmax is used.

# Softmax Function

Softmax function calculates the probabilities distribution of the event over 'n' different events. In general way of saying, this function will calculate the probabilities of each target class over all possible target classes.

## Multi-class Classification (3 classes as example)





# Pros And Cons of Softmax Activation function

## Pros

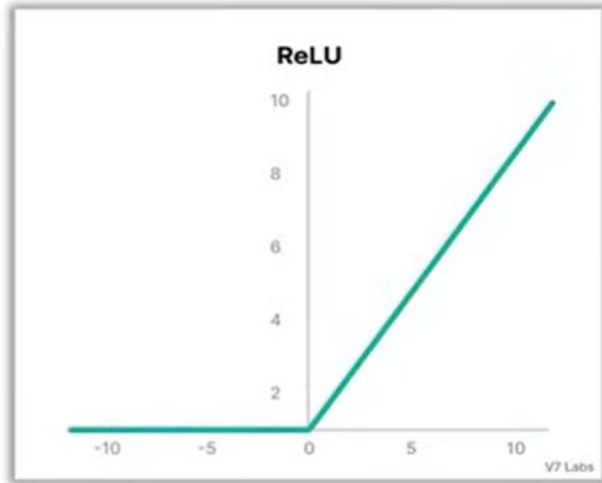
1. It mimics the one hot encoded labels better than the absolute values.
2. If we use the absolute (modulus) values we would lose information, while the exponential intrinsically takes care of this.

## Cons

1. The softmax function should not be used for multi-label classification.
2. the sigmoid function (discussed later) is preferred for multi-label classification.
3. The Softmax function should not be used for a regression task as well.

# ReLU function

ReLU Function :

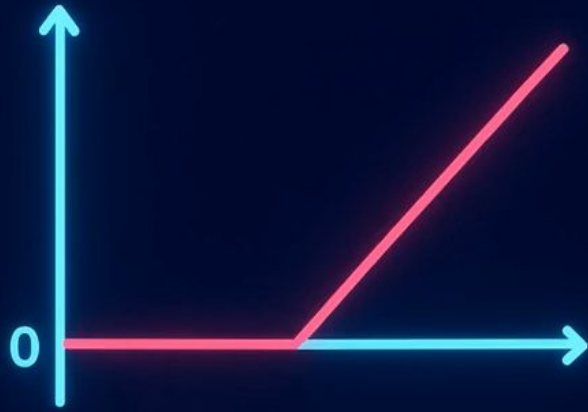


$$f(x) = \max(0, x)$$

$f(x) = x$  : Linear,  
 $f(x) = \max(0, x)$  : Non Linear, it is diffre

It's main advantage is that it avoids and rectifies vanishing gradient problem and less computationally expensive than tanh and sigmoid.

**FAST. EFFICIENT. BUT WATCH  
OUT FOR DYING NEURONS.**

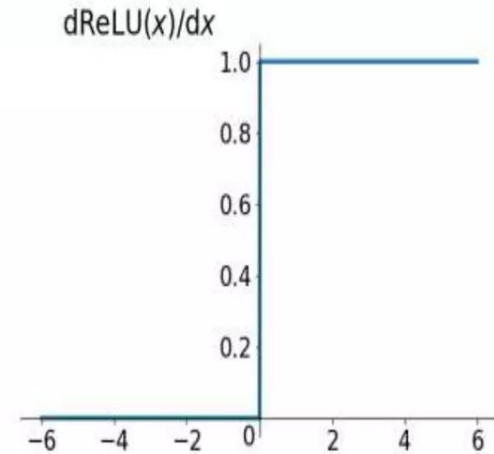
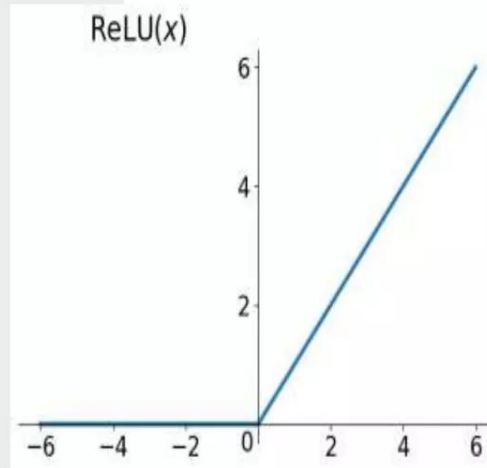


**ReLU**



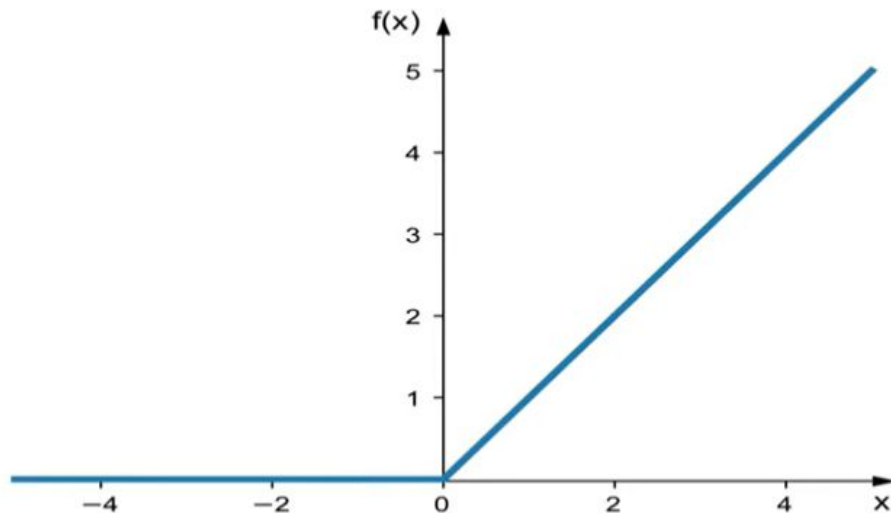
# ReLu Activation Function

The ReLU function is actually a function that takes the maximum value



# ReLU (Rectified Linear Unit)

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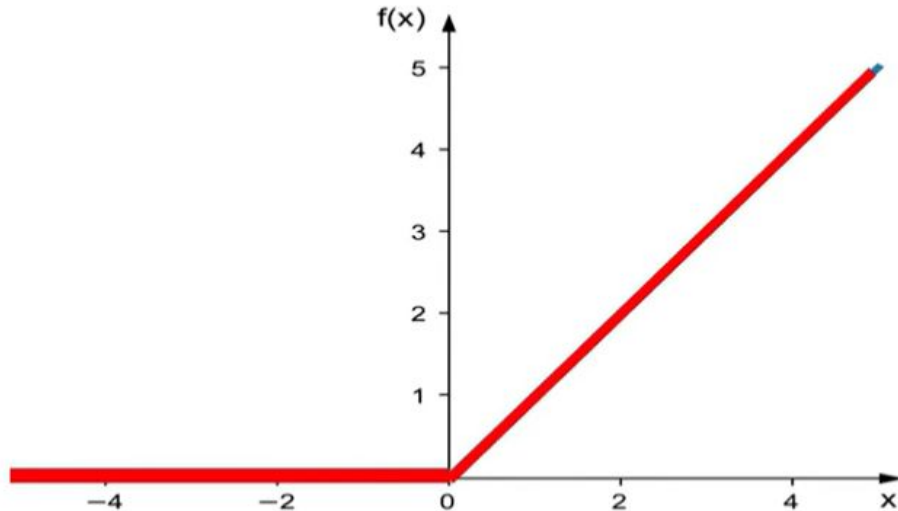
$$f(x) = \max(0, x)$$

Overcome the  
Vanishing Gradient  
Problem

$$\frac{\partial f(x)}{\partial x} = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$$

# ReLU (Rectified Linear Unit)

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Piece-wise Linear

*Advantages of both linear and non-linear property*

$$f(x) = \max(0, x)$$

Overcome the  
Vanishing Gradient  
Problem

$$\frac{\partial f(x)}{\partial x} = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$$



# Pros And Cons of ReLu function

## Pros

1. When the input is positive, there is no gradient saturation problem.
2. The calculation speed is much faster.
3. The ReLU function has only a linear relationship.
4. Whether it is forward or backward, it is much faster than sigmoid and tanh.

## Cons

1. When the input is negative, ReLU is completely inactive, which means that once a negative number is entered, ReLU will die
2. We find that the output of the ReLU function is either 0 or a positive number, which means that the ReLU function is not a 0-centric function.

# Activation Functions

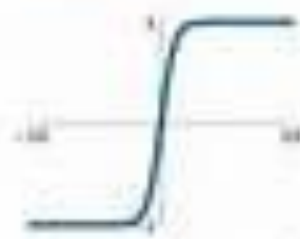
**Sigmoid**

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



**tanh**

$$\tanh(x)$$



**ReLU**

$$\max(0, x)$$



**Leaky ReLU**

$$\max(0.1x, x)$$



**Maxout**

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

**ELU**

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



# Thumb Rule for choosing right Activation Function

A few rules for choosing the activation function for output layer based on the type of prediction problem

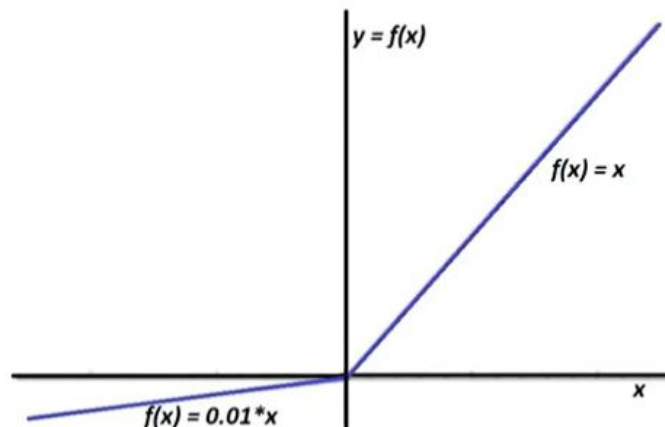
- ✓ Regression – Linear Activation Function
- ✓ Binary Classification—Sigmoid/Logistic Activation Function
- ✓ Multiclass Classification—Softmax
- ✓ Multilabel Classification—Sigmoid

The activation function used in hidden layers is typically chosen based on the type of neural network architecture :

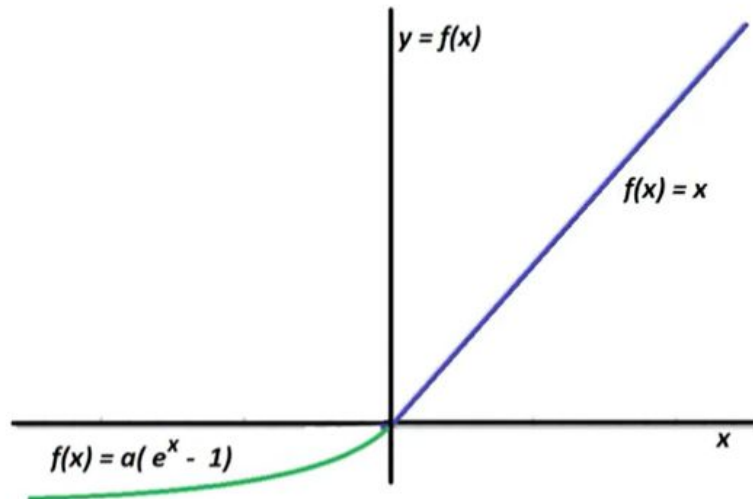
- ✓ Convolutional Neural Network (CNN): ReLU activation function.
- ✓ Recurrent Neural Network: Tanh and/or Sigmoid activation function.

# Variations of ReLU

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**Leaky ReLU**

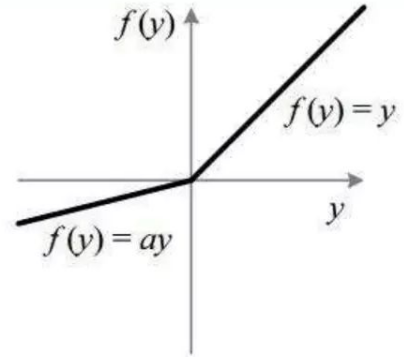
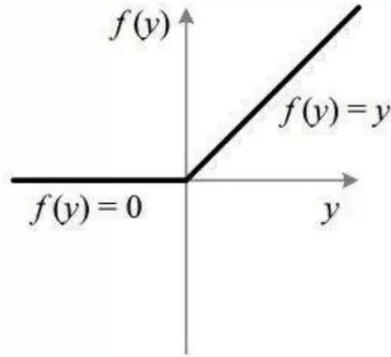


**ELU (Exponential Linear Unit)**



# Leaky ReLU Function

It is an attempt to solve the dying ReLU problem  
The leak helps to increase the range of the ReLU function.  
Usually, the value of  $a$  is 0.01 or so.





# Pros And Cons of Leaky ReLu Activation function

## Pros

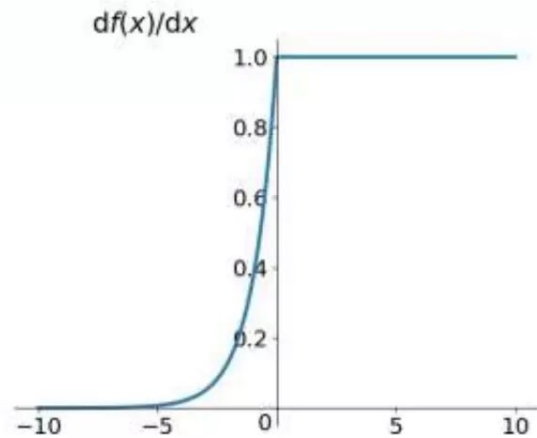
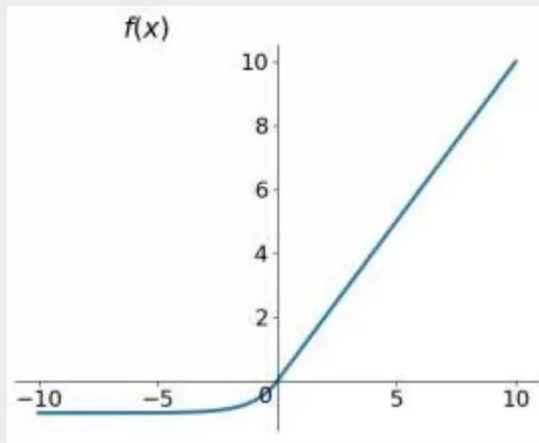
1. There will be no problems with Dead ReLU.
2. A parameter-based method, Parametric ReLU :  $f(x) = \max(\alpha x, x)$ , which  $\alpha$  can be learned from back propagation.

## Cons

1. It has not been fully proved that Leaky ReLU is always better than ReLU.

# ELU (Exponential Linear Units) function

ELU is very similar to RELU except negative inputs. They are both in identity function form for non-negative inputs. On the other hand, ELU becomes smooth slowly until its output equal to  $-\alpha$  whereas RELU sharply smooths.





## Pros And Cons of ELU Activation function

### Pros

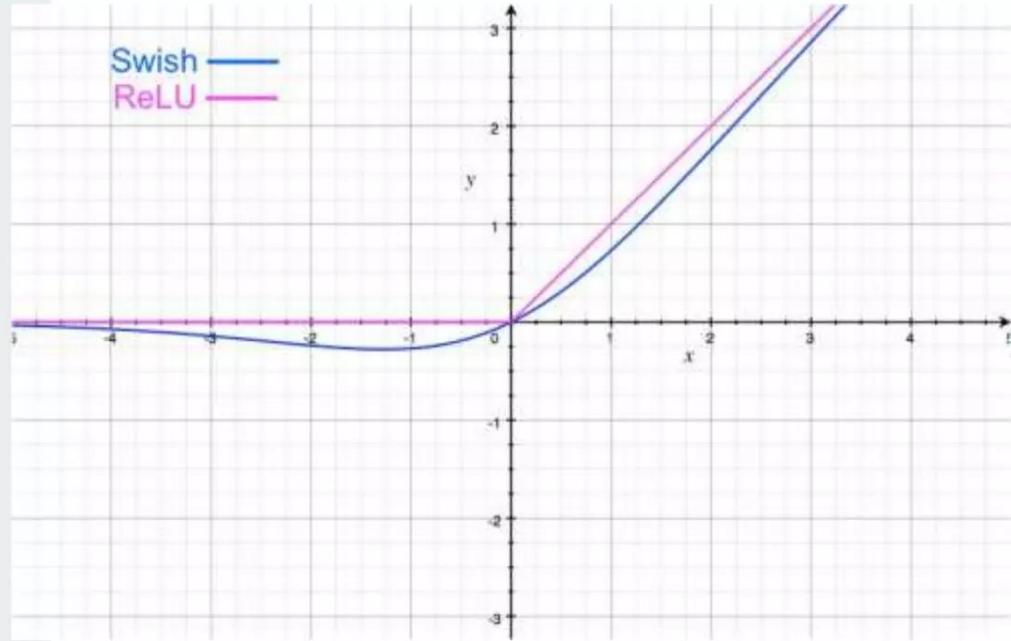
1. ELU becomes smooth slowly until its output equal to  $-\alpha$  whereas ReLU sharply smoothes.
2. ELU is a strong alternative to ReLU.
3. Unlike to ReLU, ELU can produce negative outputs.

### Cons

1. For  $x > 0$ , it can blow up the activation with the output range of  $[0, \infty]$ .

# Swish Function

Swish's design was inspired by the use of sigmoid functions for gating in LSTMs and highway networks. We use the same value for gating to simplify the gating mechanism, which is called self-gating.





# Pros And Cons of Swish Activation function

## Pros

1. No dying ReLU.
2. Increase in accuracy over ReLU
3. Outperforms ReLU in every batch size.

## Cons

1. Slightly more computationally expensive.
2. More problems with the algorithm will probably arise given time.



# Maxout Function

The Maxout activation is a generalization of the ReLU and the leaky ReLU functions. It is a learnable activation function.

The Maxout activation function is defined as follows

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$



# Pros And Cons of Maxout Activation function

## Pros

1. It is a learnable activation function.

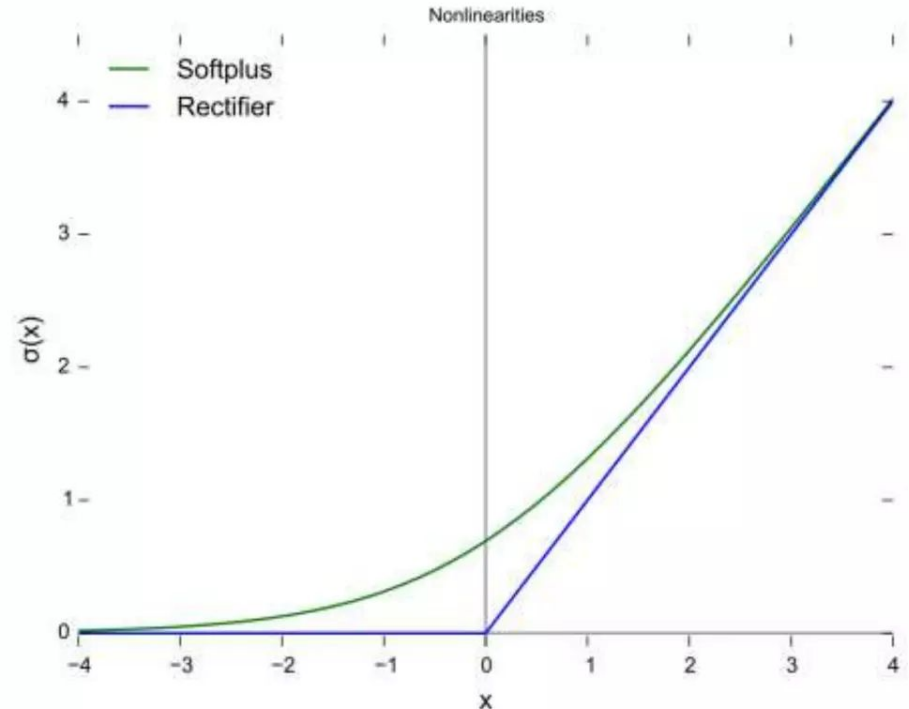
## Cons

1. It doubles the total number of parameters for each neuron and hence, a higher total number of parameters need to be trained.

# Softplus Activation Function

The softplus function is similar to the ReLU function, but it is relatively smooth. It is unilateral suppression like ReLU. It has a wide acceptance range  $(0, +\infty)$ .

Softplus function:  $f(x) = \ln(1 + \exp x)$





# Pros And Cons of Softplus Activation function

## Pros

1. It is relatively smooth.
2. It is unilateral suppression like ReLU.
3. It has a wide acceptance range (0, + $\infty$ ).

## Cons

1. Leaky ReLU is a piecewise linear function, just as for ReLU, so quick to compute. ELU has the advantage over softmax and ReLU that its mean output is closer to zero, which improves learning.

**Thank You**